NBA Cluster Analysis

Erick Anderson & Emma Johnson

University of Minnesota-Duluth

STAT 5211

Professor Melnykov

May 2, 2025

NBA Cluster Analysis

Project Overview

The objective of this cluster analysis is to group players and teams based on quality. Using NBA regular season box scores, this meant choosing variables based on an indication of a healthy player or team. We uncovered latent patterns among the data by picking variables such as shooting percentages, assists, blocks, rebounds, and others (see Table 1, NBA Data section). This required the intentional exclusion of other box score data, such as game date, which was not relevant to the objectives of these models.

As a general methodology, we combined Principal Component Analysis (PCA) with Model-Based Clustering (Scrucca et al., 2023) in R. The results inspired various interpretations of the skill levels of individual players and teams within a given season, primarily focusing on the 2023-2024 regular season. First, we focused on player models, then team models.

The Player Model. For our initial models, we used data recorded at the individual player level. While we initially considered all observations, we refined our analysis to include only players with at least 1,000 total minutes. In the NBA, not every player receives enough playing time to make a measurable impact, so this threshold helped focus on more influential players. It also reduced noise from injuries, which could skew clustering due to our use of season totals instead of per-game averages.

After further considerations, we split the player data by position. Different positions prioritize different stats. For example, centers prioritize rebounding and blocks much more than guards do. This approach helped us better capture player quality relative to others in the same role.

Overall, clustering revealed strong match-ups between players of the same position. These insights could inform practice pairings, in-game match-ups, or even guide trade considerations by highlighting comparable skills.

The Team Model. The next round of models focused on team data. With far less trial-and-error than the player-level models, the team data quickly yielded results that aligned well with the actual 2023–2024 season standings. We extended this analysis by comparing the resulting clusters to those from the 2013–2014 season to explore potential similarities. Since a lot can change from year to year, the small overlap between the two seasons showed just how much teams can shift over time.

NBA Data

In light of the current NBA Playoffs, we decided to combine our shared interest in basketball with cluster analysis. We obtained the data from the GitHub user NocturneBear (2024) through their NBA Data 2010-2024 repository. Specifically, we utilized the regular season box scores data, primarily focusing on the 2023-2024 season. The raw data had 34 features (columns) and 424,478 observations (rows). A single observation contained information about an individual player's performance during a distinct game.

Through data pre-processing, we cleaned up the large dataset to suit our analysis. This involved modifying the observational units and aggregating, as well as computing, statistics to utilize moving forward. Note that the data were split into two different data frames, one based on individual players and the other, more collective, on teams. In both cases, the data described the total performances for a single season, primarily the 2023-2024 regular season. A summary of both data frames is provided below (Table 1).

Table 1

Variables Utilized in Each Data Frame

Variable	Type	Description	**Player Data**	**Team Data**
playerID	fac	Unique player identification number	✓	
playerName	chr	Player's full name	✓	
position	chr	Center (C), forward (F), guard (G), or N/A (X)	✓	
teamID	fac	Unique team identification number		✓
teamCode	chr	3-letter team name code		✓
FGper	dbl	Field goal (two-point) percentage	✓	✓
TPper	dbl	Three-point percentage	✓	✓
FTper	dbl	Free throw percentage	✓	✓
minutes	dbl	Total minutes on the court	✓	
assists	int	Total assists	✓	✓
blocks	int	Total blocks	✓	✓
fouls	int	Total personal fouls	✓	
points	int	Total points	✓	✓
rebounds	int	Total rebounds	✓	✓
steals	int	Total steals	✓	✓
turnovers	int	Total turnovers	✓	✓

Model Synthesis

Player Data

Variables and Correlations

We began the analysis of the player data by observing the correlations between each of the numeric variables. This included the 11 variables in the third chunk of Table 1. Note that we omitted data with missing values, leaving the player data with 514 observations. A correlation plot is shown below in Figure 1.

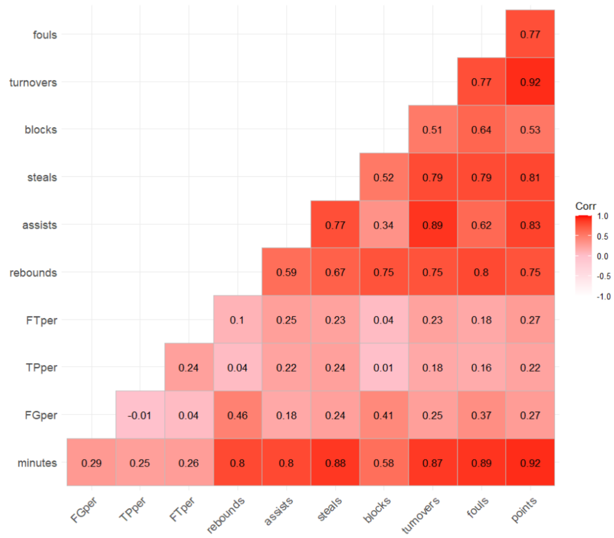

Figure 1

Player Data Correlation Plot

The correlation plot, complete with pairwise correlation coefficients, indicates that nearly all variables are moderately-to-strongly correlated with one another, aside from the statistics recorded as percentages. This is likely due to the variable minutes, suggesting that large amounts of playing time enable players to increase their points scored, assists, turnovers, etc. Field goal, three-point, and free throw percentages are more-or-less independent from minutes played. This is because they are scored based ona rate, not an aggregate score ranging from zero to infinity, as the other variables are.

Further, we can visualize the dependence relations through a pairs plot (Figure 2).

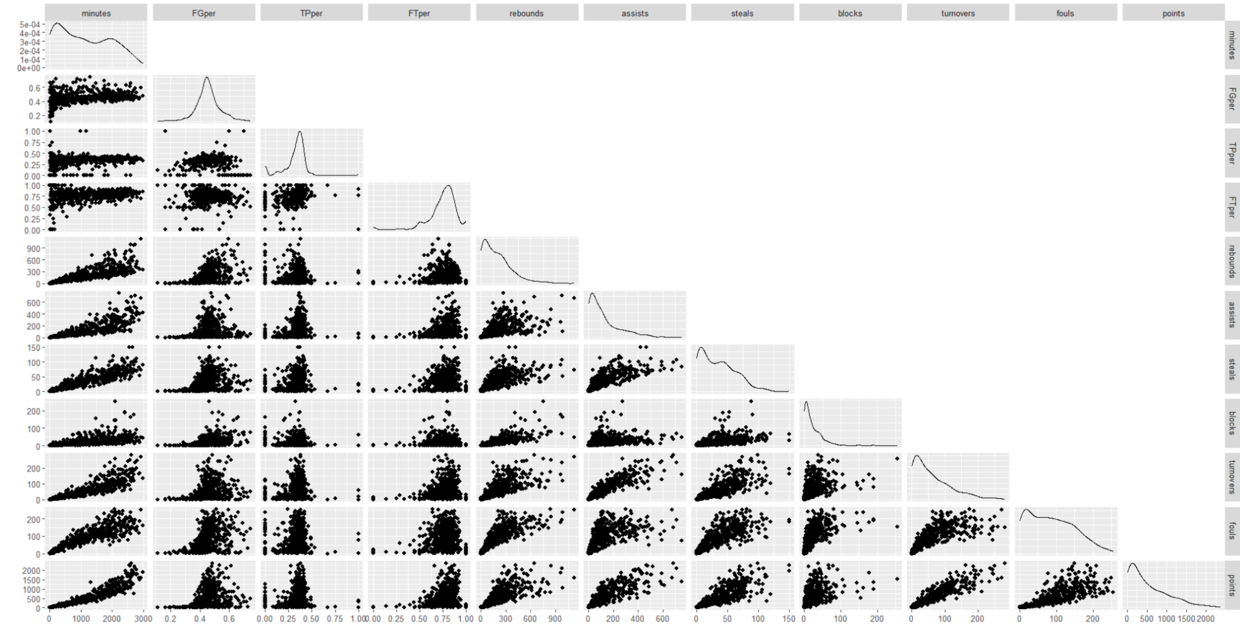

Figure 2

Player Data Pairs Plot

The pairs plot, accompanied by densities of each variable, supports the suspicion that minutes played impacts the majority of the other stats. Since many density curves exhibit right-skew, with high concentrations of observations falling in the lower range for each (non-percentage) variable, this reflects that minimal playing time is associated with lack of attempts to increase stats.

Building on these observations, we decided to move forward with the player data by omitting observations from players with negligible playing time. That is, we filtered the data to only keep the players who regularly compete throughout the season. Our method involved removing observations from players who play under 1000 minutes per regular season, averaging less than 12 minutes per game.

"Players Who Actually Play"

Now, focusing on the regular players who "actually play" throughout the season, we are working with 263 observations. This decision reflects the goal of the cluster analysis, that is, to uncover latent patterns about player strengths and weaknesses across the league. Therefore, players who lack playing time are not representative of this objective, as they consistently sit in the lowest ranges of the aggregate variable distributions in Figure 2.

Note that with this adjustment, the correlations changed slightly and the density plots adapted with less-pronounced low-range concentrations. Heavy right tails remained for many of the variables, but this is expected. After all, any professional sports league has star players who may score more, assist more, steal more, among other skills, to skew these density distributions.

Clustering Regular Players

From here, we used the R package *Mclust* (Scrucca et al., 2023) to perform model-based clustering on the regular players based on different hypotheses.

Model-Suggested Clusters

For the first run of *Mclust*, we left it up to the function to decide the number of clusters. Note that we scaled the data upon execution. The summary of this unconstrained model is displayed below.

```
----------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
----------------------------------------------------
Mclust VVE (ellipsoidal, equal orientation) model with 4 components:
 log-likelihood   n  df       BIC       ICL
      -2697.205 263 146 -6207.944 -6241.176
Clustering table:
 1  2  3  4
57 89 69 48
```

According to the unconstrained input, the model suggests 4 components that are ellipsoidal and of equal orientation (VVE). The reported BIC is 6207.944, which is minimal in the unconstrained setting with these 11 variables. Adding constraints will increase BIC.

Next, we viewed the cluster classifications for each player. With sufficient knowledge of the NBA and basketball, it should make sense that some of these players are clustered together. For example, seeing two tall guys such as Minnesota's Rudy Gobert and Milwaukee's Giannis Antetokounmpo within cluster 4 makes sense. Any grouping of players with similar positions is reassuring.

However, this is not always the case within this unconstrained model. In fact, we see in cluster 1 that Minnesota's Mike Conley, a guard, is in the same group as Denver's Kawhi Leonard, a forward. For reference, Conley stands at around 6ft tall versus Leonard, who is not only 7 inches taller, but is cited to have a wingspan longer than 7ft, according to Wikipedia. Would these two players be favorable one-on-one opponents? Probably not.

PCA Model

Perhaps dimension reduction solves these position discrepancies, leading us to utilize Principal Component Analysis (PCA) and re-run the unconstrained model. As a quick summary of this process, we found 5 Principal Components (PCs) covered around 84% of the cumulative proportion on variance across the 11 variables, thus, reducing the dimension of variables from 11 to 5. This is useful because many of the variables share similar information about the players, as presented in the preceding Variables and Correlations section.

The resulting unconstrained *Mclust* model has 6 components, described as diagonal with equal shape (VEI). The BIC value is much lower here than before, recorded at 3986.184, reflecting the decrease in parameters. Looking at the classifications, it appears that each cluster still contains a mixture of different positions.

Matching up players of opposite positions within each cluster is not overly useful, especially when a position within a cluster occurs with frequency 1. This suggests constraining the player model based on positions, instead of clustering all players together.

Players Split by Position

With these latent discoveries in the player data, we decided to run 3 separate clustering models based on each position. That is, split the data on listed position so we are left with separate data frames for centers, forwards, and guards. Note that we eliminated 8 position "X" observations as these players lacked a provided position in the data. Now, we can proceed with the remaining 255 players as described.

To solidify this choice, let's take a look at the pairs plot for the 3 positions across each variable (Figure 3).

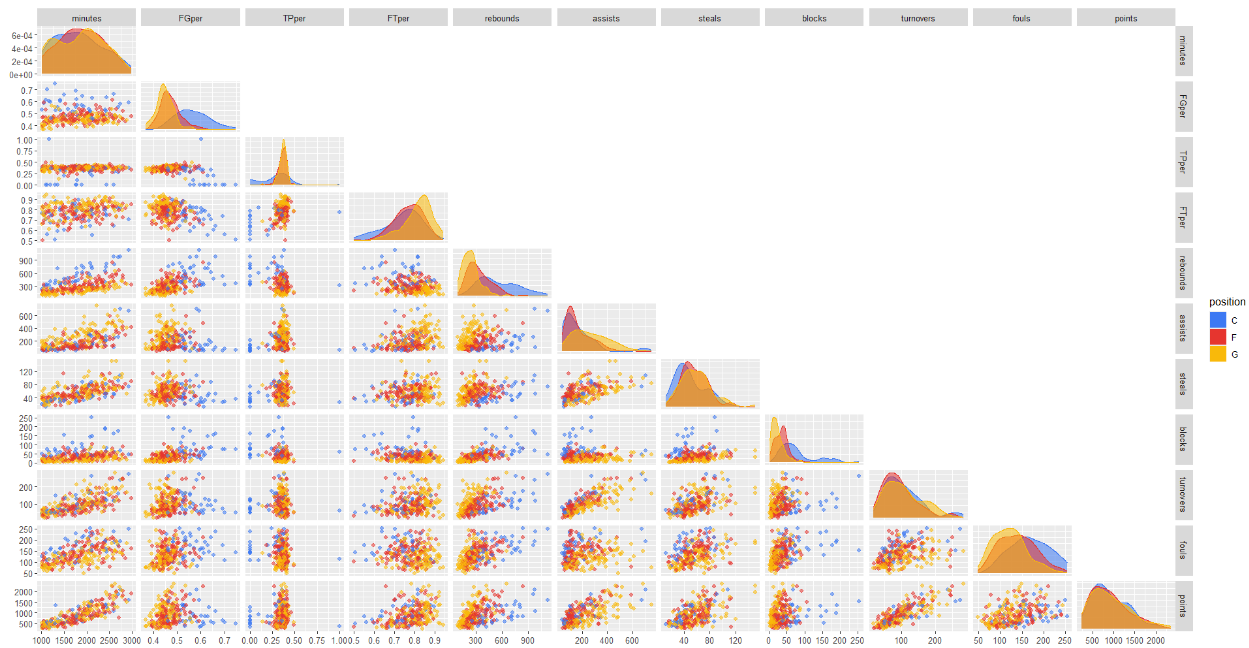

Figure 3

Player Positions Pairs Plot

Right away, we notice agreement within positions across different stats. For example, guards (G, yellow) are highly concentrated in the lower range of the density distribution for both blocks and rebounds. Guards tend to be the shorter players on the court, so recording a low degree of height-advantaged stats makes sense. Other observations include the wide range in field goal percentage for centers (C, blue), likely varying due to the high volume of shots these players typically shoot. A center is the typical offensive target towards the bottom of the hoop, meaning they may experience more contact upon shooting. Therefore, the shot may be closer, suggesting the ball is more likely to go in, but also more likely to be contended.

For each of the position-based clustering models, we continue using *Mclust* with PCA. This ensures dimension-reduction among the input variables, as explained in the previous section. Also, we constrained each model with 3 clusters, aiming to represent 3 talent levels. The motivation for this was the idea of Varsity, Junior Varsity, and B-Squad teams.

Centers (41 players)

PCA. We used 4 principal components for the centers data, which cumulatively explain around 83% of the variation across the variables. The matrix of variable loadings under the 4 PCs is provided below, followed by an interpretation.

	PC1	PC2	PC3	PC4
minutes	0.39536064	-0.08213685	0.152319469	-0.04004470
FGper	-0.12266102	-0.51172630	-0.161434227	0.27087600
TPper	0.04822617	0.53492463	0.021987841	-0.24420083
FTper	0.10516056	0.55650347	0.066866660	0.26593545
rebounds	0.36290954	-0.28187531	0.001224589	0.15809361
assists	0.37237388	0.04822810	-0.406662862	0.05842069
steals	0.36485376	-0.02565519	-0.104810448	0.06234652
blocks	0.16904351	-0.14843672	0.849720490	-0.06458647
turnovers	0.40052811	0.02193222	-0.144744835	0.05429220
fouls	0.26602040	-0.14977049	-0.113139097	-0.79947606
points	0.38988523	0.09036448	0.116813134	0.34072310

PC1 is generally inclusive of all variables, suggesting commonalities in all provided stats, with the exception of a slightly negatively loaded FGper. PC2 contrasts FGper with TPper and FTper, highlighting scoring range. PC3 primarily contrasts assists with blocks, an offensive/defensive comparison, and PC4 contrasts fouls with points, the differentiating between penalties and offensive scoring, respectively.

Centers are typically the "big guys” in basketball, so it makes sense that these PCs pick up on the actions of this type of player. A center is typically playing close to the basket, so they are more likely to have a higher FGper than TPper, as contrasted in PC2. Also, they tend to have many blocks, as highlighted in PC3.

Mclust. Now, lets take a look at a model utilizing these 4 PCs with the accompanied classifications (Table 2).

```
----------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
----------------------------------------------------
Mclust EVE (ellipsoidal, equal volume and orientation) model with 3 components:
 log-likelihood  n df       BIC       ICL
      -241.1538 41 30 -593.7148 -599.8501
Clustering table:
 1  2  3
19 12 10
```

Table 2

Centers cluster classifications

Cluster 1	...	Cluster 2	Cluster 3
Al Horford	Grant Williams	Alperen Sengun	Andre Drummond
Daniel Theis	Isaiah Hartenstein	Anthony Davis	Clint Capela
Deandre Ayton	Jalen Duren	Bam Adebayo	Dereck Lively II
Drew Eubanks	Jarrett Allen	Brook Lopez	James Wiseman
Duop Reath	Joel Embiid	Chet Holmgren	Kevon Looney
Jonas Valanciunas	Jusuf Nurkic	Domantas Sabonis	Nic Claxton
Lauri Markkanen	Marvin Bagley III	Kelly Olynyk	Nick Richards
Moritz Wagner	Onyeka Okongwu	Kristaps Porzingis	Rudy Gobert
Paul Reed	Wendell Carter Jr.	Myles Turner	Trayce Jackson-Davis
Zach Collins		Nikola Jokic	Walker Kessler
		Nikola Vucevic	
		Victor Wembanyama	

One important note is that this split based on position is much more digestible to interpret. Instead of drawing conclusions based on a list spanning over 250 names, we now look at a subset that is easier to seek comparisons on. Right away, cluster 2 groups Kristaps Porzingis and Nikola Jokic together, a pair of players that are regularly compared in the NBA. Therefore, a model-based suggestion to match up the 2 players appears justified.

The following pairs plot provides a visual representation of these clusters (Figure 4).

Figure 4

Clustered Centers

There are many facets in this visual, but consider building on the previously drawn observation. If we look at cluster 2 (red), the one with Porzingis and Jokic, we see that they fall within the highest range for both minutes and points, naturally. They also have the highest FTper. We may consider this group to be "Varsity" out of the 3 clusters, depending on the bias of the interpreter.

Forwards (106 players)

PCA. With the forward position data, 5 PCs make up around 85% of the cumulative proportion of variation. The matrix of variable loadings for these PCs is found below.

```
                PC1         PC2         PC3         PC4         PC5
minutes   -0.39686494 -0.106638951 -0.23335344  0.018489448  0.03110136
FGper     -0.18896428  0.007474829  0.72036442  0.257856199 -0.02320575
TPper      0.05087411 -0.596598098  0.26385191  0.422819298 -0.26938670
FTper     -0.09060384 -0.620553822 -0.29696944  0.139511636  0.32139051
rebounds  -0.35217143  0.179893108  0.14486330  0.007759175  0.38015176
assists   -0.36338523 -0.110609967  0.15421345 -0.433638650 -0.15768746
steals    -0.29854497 -0.021501438 -0.37087840  0.015081727 -0.63518635
blocks    -0.24461883  0.357294095 -0.01743460  0.447209543 -0.33280396
turnovers -0.39551660 -0.041007961  0.12355665 -0.270583745  0.07941406
fouls     -0.25876009  0.202304500 -0.26020910  0.516644703  0.36142415
points    -0.40934375 -0.179526275  0.05670684 -0.080404204  0.04580692
```

A similar interpretation holds for PC1 as observed with the centers data (see previous). Here, PC2 groups FTper and TPper in contrast to blocks. PC3 mainly contrasts FGper with FTper, blocks, and fouls, making up a variable that generally describes points scored during running time versus stopping time. PC4 differentiates TPper, blocks, and fouls from both assists and turnovers, contrasting aggressive actions with ball release. Finally, PC5 contrasts both blocks and steals from rebounds, fouls, and FTper, differentiating defense moves from other aggressive behaviors.

Forwards are typically the most versatile players. They exhibit a strong shooting range and tend to fall in the mid-range of heights within the NBA. They are aggressive on both sides of the court, which the principal components seem to pick up on.

Mclust. Moving along, the 5 PCs plugged into *Mclust* yields the following results.

```
----------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
----------------------------------------------------
Mclust VEI (diagonal, equal shape) model with 3 components:
 log-likelihood   n df       BIC       ICL
       -819.792 106 24 -1751.507 -1790.596
Clustering table:
 1  2  3
53 35 18
```

Table 3

Forwards cluster classifications

Cluster 1	...	Cluster 2	...	Cluster 3
Aaron Gordon	Grayson Allen	Andrew Wiggins	Franz Wagner	Amir Coffey
Aaron Nesmith	Haywood Highsmith	Bennedict Mathurin	Georges Niang	Bojan Bogdanovic
Ausar Thompson	Isaac Okoro	Bobby Portis	Isaiah Stewart	Cameron Johnson
Bilal Coulibaly	Jaden McDaniels	Brandon Ingram	Jabari Smith Jr.	Cedi Osman
Buddy Hield	Jae'Sean Tate	Brandon Miller	Jabari Walker	Dean Wade
Cason Wallace	Jalen Johnson	Dario Saric	Jaime Jaquez Jr.	GG Jackson
Dante Exum	Jalen Smith	De'Andre Hunter	Jeff Green	Gordon Hayward
DeMar DeRozan	Jalen Williams	Deni Avdija	Jeremy Sochan	Harrison Barnes
Derrick Jones Jr.	Jaren Jackson Jr.	Dillon Brooks	Jonathan Kuminga	Jae Crowder
Donte DiVincenzo	Jaylen Brown	Josh Hart	Julius Randle	Jerami Grant
Dorian Finney-Smith	Jayson Tatum	Karl-Anthony Towns	Keldon Johnson	Julian Champagnie
Draymond Green	Jimmy Butler	Khris Middleton	Kyle Kuzma	Moses Moody
Evan Mobley	John Collins	Max Strus	Michael Porter Jr.	Naji Marshall
Giannis Antetokounmpo	Kawhi Leonard	Miles Bridges	Miles Bridges	Patrick Williams
Kelly Oubre Jr.	Keegan Murray	Pascal Siakam	RJ Barrett	Simone Fontecchio
Kenrich Williams	Kevin Durant	Saddiq Bey	Sam Hauser	Torrey Craig
Kris Murray	Kyle Anderson	Terance Mann	Tobias Harris	Trey Murphy III
LeBron James	Luguentz Dort	Toumani Camara	Ziaire Williams	Vince Williams Jr.
Matisse Thybulle	Naz Reid			
Mikal Bridges	Nicolas Batum			
Obi Toppin	Ochai Agbaji			
OG Anunoby	P.J. Washington			
Paolo Banchero	Paul George			
Peyton Watson	Precious Achiuwa			
Royce O'Neale	Rui Hachimura			
Santi Aldama	Taurean Prince			
Zion Williamson				

Something worth noting is that these clusters vary in size more than the clusters found with the center position players. For the forwards, cluster 1 is about 3 times the size of cluster 3. If we are still basing the 3 clusters in terms of talent level, this just means we have a non-uniform distribution of our hypothesized skill groups, which is no cause for concern. Adjusting the number of clusters may make the group sizes more-or-less equal, but we will continue with 3 clusters for the sake of consistency in interpretations.

As an interpretation example, notice how Julius Randle and Karl-Anthony Towns appear in cluster 2. Minnesota traded Towns for Randle (and others) in the fall of 2024. While many MN fans were upset with this trade, it appears that the model has picked up similarities between the two players, which could support the trade decision.

Another look at the classifications reveals that many of the decades-long stars fall into cluster 1, with names like Kevin Durant, Kawhi Leonard, Jimmy Butler, Paul George, and LeBron James. The NBA veterans are accompanied by some rising-in-status players, such as Minnesota's beloved 6th Man of the Year, Naz Reid. One way to utilize this observation is to pair up newer players with seasoned ones. That way, the new stars can improve upon their game with the help of their mentors so they, too, can become one of the greats.

Figure 5

Clustered Forwards

It is not clear whether cluster 1 (blue) or 2 (red) is "better", but it may be safe to assume that cluster 3 (yellow) represents the "B-Squad" according to this model. That is, cluster 3 players fall in the lower range for minutes, thus, having lower assists, steals, blocks, etc. One should be cautious about drawing conclusions based solely on percentage variables. Cluster 3 has a strong high-range TPper concentration, yet these players see the line less as a consequence of a lack of playing time.

Guards (108 players)

PCA. To wrap up the position analysis, the guards data proceeds with 5 principal components. These 5 PCs, with loadings captured below, provide around 85% of the cumulative proportion of variance.

	PC1	PC2	PC3	PC4	PC5
minutes	0.40041740	0.11766377	0.01396406	0.10178426	0.13081324
FGper	0.19261928	-0.16172574	-0.55732693	-0.51776693	0.09350152
TPper	0.08054927	0.44181187	-0.63835779	-0.03987644	0.04002971
FTper	0.10597145	0.67273659	0.03338250	0.22354054	-0.38169367
rebounds	0.33489126	-0.28668262	0.12574661	-0.19018598	-0.27019334
assists	0.35074791	0.12989086	0.20880462	-0.30137483	-0.06391794
steals	0.33801100	-0.23113035	-0.16871637	0.13017345	0.02966163
blocks	0.26446554	-0.30813497	-0.21473169	0.37272061	-0.65017451
turnovers	0.36632039	0.08967020	0.33978552	-0.16157983	0.13942723
fouls	0.28864454	-0.09987541	-0.10604862	0.58908534	0.54729150
points	0.38466814	0.21629835	0.14103067	-0.11833856	0.08478399

PC1, once more, describes the variables in unison. PC2 contrasts TPper and FTper with blocks and rebounds, describing scoring range versus gaining or denying possession near the basket. PC3 contrasts FGper and TPper with turnovers, that is, point acquisition with loss of the ball. PC4 differentiates FGper and assists from fouls and blocks, specifically offensive versus defensive stats. Lastly, PC5 contrasts blocks with fouls, describing the subtle differences in contact when trying to defend the ball.

Guards are typically the shortest men on the court, who specialize in ball handling and play-making. They can be described as "scrappy", both making plays happen on offense and stopping them on defense. So, PC2 describes the shooting range of these players well, along with a lack of under-the-basket presence. Further, PC4 and PC5 describe ball control, closely resembling attempts to snag the ball away on defense. Specifically, PC5 illustrates the moment where an attempted steal or block can turn into a foul if executed poorly.

Mclust. With the 5 PCs, *Mclust* creates the final position model.

```
---------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
---------------------------------------------------
Mclust VEI (diagonal, equal shape) model with 3 components:
 log-likelihood   n df      BIC       ICL
      -826.6494 108 24 -1765.67 -1804.946
Clustering table:
 1  2  3
43 52 13
```

Table 4

Guards cluster classifications

Cluster 1	...	Cluster 2	...	Cluster 3
Aaron Holiday	Malaki Branham	Aaron Wiggins	Fred VanVleet	Austin Reaves
Alec Burks	Malcolm Brogdon	Alex Caruso	Herbert Jones	Cade Cunningham
Anfernee Simons	Marcus Sasser	Amen Thompson	Jaden Ivey	Coby White
Bruce Brown	Miles McBride	Andrew Nembhard	Jalen Green	Collin Sexton
Caleb Martin	Pat Connaughton	Anthony Black	Jalen Suggs	D'Angelo Russell
Cam Thomas	Patrick Beverley	Anthony Edwards	Jamal Murray	Damian Lillard
Cameron Payne	Payton Pritchard	Ayo Dosunmu	James Harden	Darius Garland
Chris Paul	Quentin Grimes	Bogdan Bogdanovic	John Konchar	Dennis Schroder
Dalano Banton	Reggie Jackson	Bradley Beal	Jordan Goodwin	Devin Booker
Davion Mitchell	Sam Merrill	Brandin Podziemski	Jordan Poole	Jalen Brunson
De'Anthony Melton	Shaedon Sharpe	Caris LeVert	Josh Giddey	Stephen Curry
Desmond Bane	Spencer Dinwiddie	Christian Braun	Josh Green	Trae Young
Duncan Robinson	Talen Horton-Tucker	CJ McCollum	Jrue Holiday	Tyrese Maxey
Eric Gordon	Terry Rozier	Corey Kispert	Kentavious Caldwell-Pope	
Gary Harris	Tim Hardaway Jr.	David Roddy	Killian Hayes	
Gary Trent Jr.	Tyler Herro	De'Aaron Fox	Kris Dunn	
Gradey Dick	Vasilije Micic	Dejounte Murray	Kyrie Irving	
Immanuel Quickley	Isaiah Joe	Dennis Smith Jr.	Luka Doncic	
Jordan Clarkson	Jordan Hawkins	Derrick White	Malik Beasley	
Josh Richardson	Kevin Huerter	Devin Vassell	Mike Conley	
Klay Thompson	Keyonte George	Donovan Mitchell	Nickeil Alexander-Walker	
Kyle Lowry		Dyson Daniels	Norman Powell	
		Russell Westbrook	Scoot Henderson	
		Scottie Barnes	Shai Gilgeous-Alexander	
		T.J. McConnell	Tre Jones	
		Tyrese Haliburton	Tyus Jones	

The resulting model provides a diagonal, equal shape (VEI) fit with unequal cluster observation sizes. Cluster 2 contains a few big names, such as Russel Westbrook, James Harden, and Kyrie Irving. It also has Minnesota's Anthony Edwards and LA's Luka Doncic, who recently faced off in the NBA playoffs (as of late April 2025). These two players exhibited an intense match-up, providing evidence to the clustering.

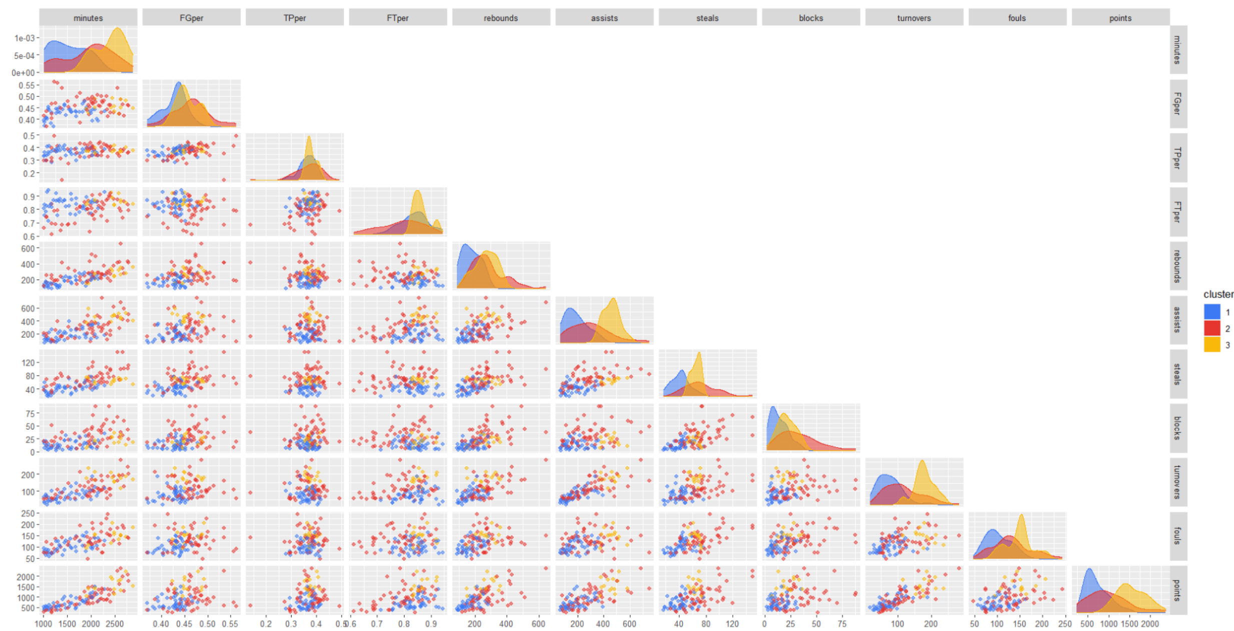

Figure 6

Clustered Guards

Cluster 2 (red) spreads wide across most of the variables, whereas cluster 3 (yellow) concentrates in the higher range for many plots. Cluster 3 may describe talented players who risk foul trouble, as seen by the higher concentration in the upper range of fouls. Depending on the severity and time of the game, these players may be risky to have on the court, yet, their payoff is high in terms of consistency. Clusters 2 and 3 share the right upper tails of many of the variable density distributions, so choosing which is "JV" or "Varsity" could be a toss-up. However, cluster 1 most likely represents the hypothesized "B-Squad" group, as they take on the left-ends of the distributions.

Team Data

A similar process takes place for the team data as seen in the Player Data section. That is, we utilize PCA to reduce the number of variables in *Mclust*. We will begin the team analysis by taking a look at the 2023-2024 season.

Variables and Correlations

First, note the 9 variables in Table 1.

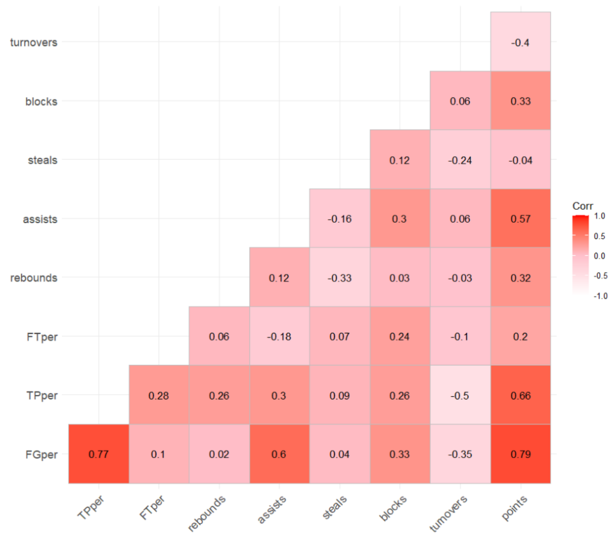

Figure 7

Team Data Correlation Plot

Again, we have FGper, TPper, FTper as percentage (rate) variables, with the rest being aggregate scores. Checking for correlations among variables, we conclude that most variables are weakly correlated. However, FGper, TPper, and assists share moderate-to-strong pairwise correlation coefficients. This is expected, as they all describe the typical scoring breakdown. Additionally, turnovers and TPper have a moderate negative correlation, which PCA may highlight.

As another visual, a pairs plot of the team data spread across the variables is shown below (Figure 8).

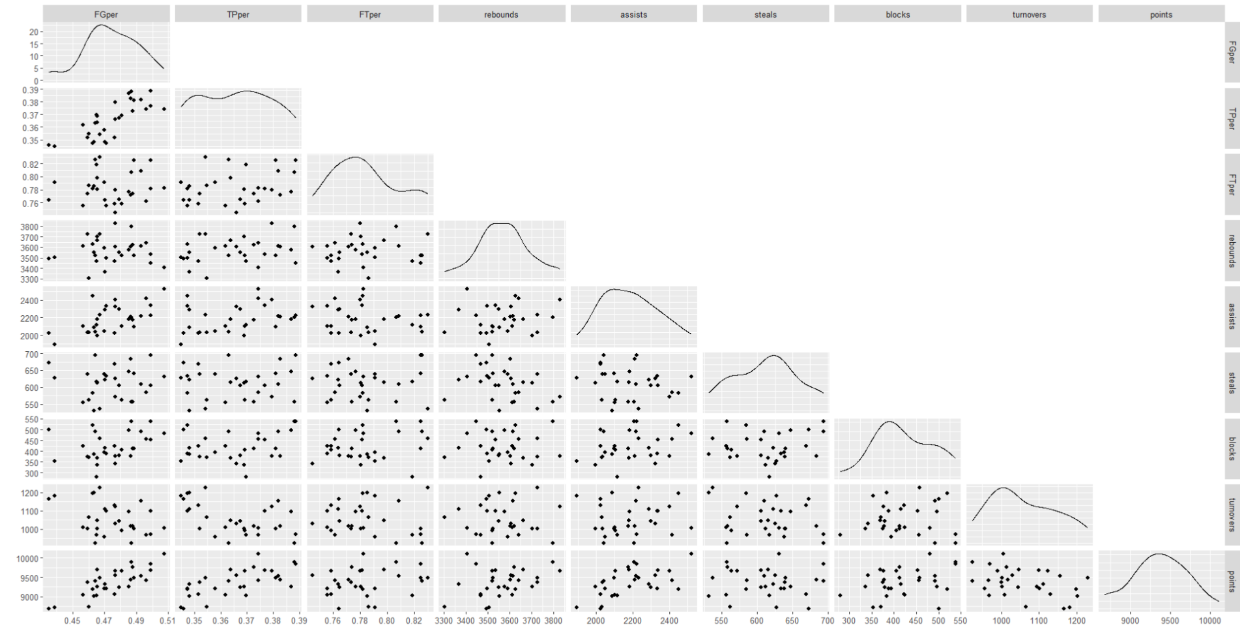

Figure 8

Team Data Pairs Plot

The density plots along the diagonal show that many of the distributions form a normal-like or uniform-like curve. This is quite different than the skewness seen in the player data plots. There are only 30 observations in the 2023-2024 team data, much less than in the player dataset. The observation that the distributions are uniform/normal-like suggests that teams are more-or-less the same as compared to player differences, which vary more.

PCA

4 PCs provided around 80% of the cumulative proportion of variance.

	PC1	PC2	PC3	PC4
FGper	0.49646604	-0.00352295	-0.19076209	0.13362706
TPper	0.47071613	-0.15208068	0.18400146	0.05110026
FTper	0.13632576	-0.30871832	0.33639381	-0.64152777
rebounds	0.15193666	0.38254996	0.58926545	-0.09763226
assists	0.33509065	0.41031823	-0.40861798	0.08931702
steals	0.01532103	-0.61564259	-0.31076077	0.02758525
blocks	0.24772317	-0.01106338	-0.34146639	-0.60359585
turnovers	-0.25877889	0.42502656	-0.29525529	-0.42866067
points	0.49942424	0.08402831	0.04519769	0.04032660

PC1 contrasts productive stats with turnovers, the only obstructive stat in the data. PC2 describes events that typically occur when the ball is in the air, that is, turnovers, rebounds, and assists, versus FTper and steals, the latter associated with extra scoring opportunities. PC3 contrasts rebounds with assists, distinguishing random ball movement with controlled movement, and PC4 groups FTper, blocks, and turnovers, perhaps high-intensity stats.

Mclust

Unconstrained Model. Next, we fit a model free of additional constraints to see how many clusters it suggests.

```
----------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
----------------------------------------------------
Mclust EEV (ellipsoidal, equal volume and shape) model with 8 components:
 log-likelihood  n df       BIC      ICL
       46.50164 30 91 -216.5057 -216.506
Clustering table:
1 2 3 4 5 6 7 8
5 4 4 3 4 4 3 3
```

Table 5

Team cluster classifications (unconstrained 2023-2024)

Cluster 1	2	3	4	5	6	7	8
ATL	CLE	CHI	LAC	IND	CHA	DET	BKN
BOS	DEN	HOU	OKC	LAL	MEM	SAS	TOR
DAL	NOP	MIA	PHI	MIN	ORL	UTA	WAS
GSW	SAC	NYK		PHX	POR		
MIL							

Under the unconstrained model, Mclust suggests 8 ellipsoidal, equal volume and shape (EEV) clusters with a BIC of 216.5057. Interestingly enough, the 16 teams who made it to the playoffs in 2024 are scattered across 6 of the 8 clusters. Clusters 7 and 8 contain no teams who made it to the playoffs, whereas clusters 4 and 5 contain *only* teams who made it. Overall, 8 clusters may be too many for our purposes, so let's implement a constraint on the clusters moving forward.

Constrained with 5 Components. We chose to reduce the clusters down to 5 to see which teams were alike. The choice of 5 is mostly random, but out of 30 teams, it seemed to create somewhat equal cluster sizes. The results are as follows.

```
----------------------------------------------------
Gaussian finite mixture model fitted by EM algorithm
----------------------------------------------------
Mclust VII (spherical, varying volume) model with 5 components:
 log-likelihood  n df       BIC       ICL
      -179.2083 30 29 -457.0513 -461.1454
Clustering table:
1 2 3 4 5
2 8 4 8 8
```

Table 6

Team cluster classifications (constrained 2023-2024)

Cluster 1	2	3	4	5
SAS	ATL	CHI	BOS	BKN
UTA	CLE	HOU	IND	CHA
	DAL	MIA	LAC	DET
	DEN	NYK	LAL	MEM
	GSW		MIN	ORL
	MIL		OKC	POR
	NOP		PHI	TOR
	SAC		PHX	WAS

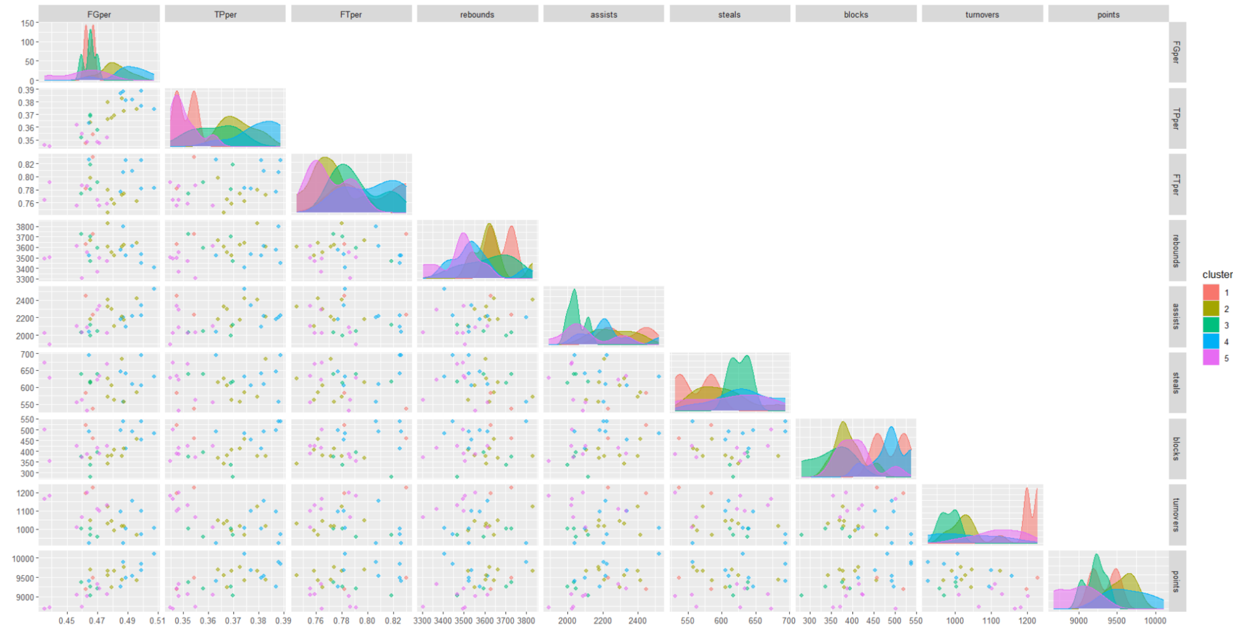

Figure 9

Clustered Teams

Going cluster-by-cluster, we draw some observations about the different skill sets of the teams. First, cluster 1, including the San Antonio Spurs and the Utah Jazz, displays the highest rate of turnovers by a considerable margin. Further, they fall in the lower range for steals, suggesting that the teams lacked ball control and extra possessions. Cluster 2, with the Milwaukee Mavericks, Denver Nuggets, and Cleveland Cavaliers, to name a few, were consistent in high scoring games, i.e. points, as well as moderate rebounds throughout the season. The Miami Heat, New York Knicks, Chicago Bulls, and Houston Rockets in cluster 3 displayed low turnovers, blocks, and assists, with a moderate-to-high amount of steals. This highlights their defensive and offensive specialties and weaknesses, as compared to the other teams. Cluster 4 has the Minnesota Timberwolves, Boston Celtics, and Indiana Pacers, among others, which were 3 out of the 4 teams who played in the 3rd round of the 2024 NBA playoffs. These teams share high shooting percentages and points, accompanied by plenty of blocks throughout the regular season. Finally, cluster 5, with 1 out of the 8 teams making it to the playoffs, i.e. the Orlando Magic, scored low in points throughout the season, paired with low rebounds and assists and somewhat high turnovers. These teams could be viewed as lacking aggression during the 2023-2024 season.

Throwback Season

Much changes year-to-year for each team in the NBA, let alone over the course of several seasons. So, we decided to compare the previous model with a different season selection. This final analysis compares cluster similarities between the 2023–2024 and 2013–2014 team data.

Again, we applied PCA, but discussion of this step is omitted because the results closely align with those discussed earlier. Instead, we will skip ahead and present the classification results from the *Mclust* model.

Table 7

Team cluster classifications (2013-2014)

Cluster 1	2	3	4	5
ATL	BOS	CHA	DEN	DET
BKN	CHI	MEM	GSW	PHI
DAL	CLE	NOP	HOU	
LAC	IND	NYK	LAL	
MIA	MIL		OKC	
MIN	ORL		PHX	
WAS	SAC		POR	
	TOR		SAS	
	UTA			

As expected, the 2013-2014 clusters are quite different from the 2023-2024 ones. In fact, the maximum similarity is that 3 teams are within the same cluster for both seasons, that is, the LA Lakers, Oklahoma City Thunder, and Phoenix Suns. Any other cluster may have a pair of 2 teams similar to the previous clustering, at most. So, clustering may only be useful in this data to support conclusions on a small time scale, much smaller than a decade difference in time.

Model Applications

These models succeed in clustering players with similar ranges in points, assists, rebounds, etc. This could potentially provide insight for teams and coaches about who should practice together, what defensive sets to form, or what skills the team or player possesses or lacks.

The clusters to which the teams are assigned could provide insight into the current team's quality. Using data from previous seasons, one could observe which teams had previously been in similar situations to examine how they improved. For example, the Washington Wizards and Detroit ended up in Cluster 5 in the 2023-2024 constrained model. Interpretation suggests this as the bottom of the league, as all teams in this group had below 0.500 winning percentage. However, for the 2024-2025 season, Detroit has improved significantly, whereas Washington would likely continue to sit in the cluster of teams ranked low in the league. The general manager of Washington could compare their team with the Detroit Pistons of 2023-2024 to inspect their changes in stats. This could guide drafting decisions and choosing which free agents to sign.

Teams could even experiment with combining the models. For example, the Wizards could compare their current roster with the similarly-clustered teams from the past. Then, they could focus on examining teams that improved during the present season. Next, the GM could use the player model to see what issues were addressed by the teams that improved, or even examine the teams that fell into the same cluster as them. This could provide valuable insight on high-impact players in the current NBA landscape.

Further, we could easily adapt our model to fit college basketball players, thus, assisting NBA teams with a drafting tool. College box scores are readily available, and a similar clustering procedure could help suggest the caliber of such prospects and what holes they could fill, all determined from the NBA teams and players models.

Discussion

For the most part, the models performed as expected, synchronous with prior knowledge of cluster analysis in professional sports. As explored, the models grouped players of similar skill levels together. For example, when performing the clustering on forward positions, Giannis Antetokounmpo and LeBron James were clustered together. LeBron is considered to be one of the greatest players in history and Giannis, as NBA insiders suggest, is a candidate for a top 10 best player of all time.

However, the clusters were often larger than initially expected. We assumed that the

"Varsity" cluster, that is, the one with the most promising talent, would be significantly smaller than the other two clusters. The models refuted this expectation, as both the lower-talent level and middle-talent level clusters were consistently small for the forwards and guards clustering. Specifically, the forwards top-talent cluster contained 53 players, the middle-talent contained 35, and the lower-talent cluster contained 18. So, the top-talent cluster is equal in size to the middle and bottom-talent clusters combined. This implies that if a player is chosen at random, it is a 50/50 likelihood that they'd be considered top-talent or either of the the lower options. That is extremely interesting as our initial expectations assumed much larger proportions of players below top-talent.

The centers broke this pattern with the top-talent making up the second largest of the 3 clusters. An explanation for this, perhaps, is that the number of centers is small compared to the other positions. Along this consideration, it may be difficult to distinguish a top-talent level for this position. Biology supports this notion, as Centers are the tallest players on the floor. The average height of a NBA center is 6' 10", a height that makes up less than 1% of the world population. So, the uncommon height attribute could compensate for the talent level necessary to get drafted in the league, thus, a lower threshold.

Further, there are a couple of immediate changes we would make to improve the variable selection for the cluster analysis. For one, the shooting percentage variables require some re-evaluation. Especially in the case of observing position differences or team differences, it may be more useful to consider shooting variables as aggregate scores. That way, teams or players who make many three-pointers will stand apart from those who don't. For example, say a player only shot one three-pointer all season. Then, their three-point shooting average (i.e. either 0 or 1) would be negligible when compared with others who shoot threes more often. There is a good chance that aggregating shots instead of averaging them would change some model results.

With the knowledge gained, there are few changes to make to the models going forward. One, potential change is to look at averages rather than totals. This would eliminate the need to ignore players with injuries, low playing time, etc. This would increase usability for a potential client as season averages are easier to understand as opposed to total. Looking at averages would also make it easier to understand the roll of bench players, as a team maybe looking to add 10 points per game and its not entirely clear what players would fit that mold based on the current model. In a similar vein, adding salary to the player model would be a good addition. The NBA has a salary cap for players meaning teams would want to maximize skill for salary. A cluster may provide the insight of a good value player. These would be players that have the top skill level but the salary doesn't necessarily match.

Apart from cluster size and variable modifications, the models seemed to meet our expectations. Since there are no right or wrong clustering classifications in this exploratory analysis, the methodology and procedure performed well, once we faced the more obvious limitations such as addressing playing time for bench players. After all, we sought to uncover latent patterns in the data, for which PCA and model-based clustering work wonderfully together. Therefore, the models aligned with our expectations by clustering players based on interpretations of talent.

References

NocturneBear. (2024). Nba data 2010-2024.

Scrucca, L., Fraley, C., Murphy, T. B., & Raftery, A. E. (2023). *Model-based clustering, classification, and density estimation using mclust in R*. Chapman; Hall/CRC. https://doi.org/10.1201/9781003277965